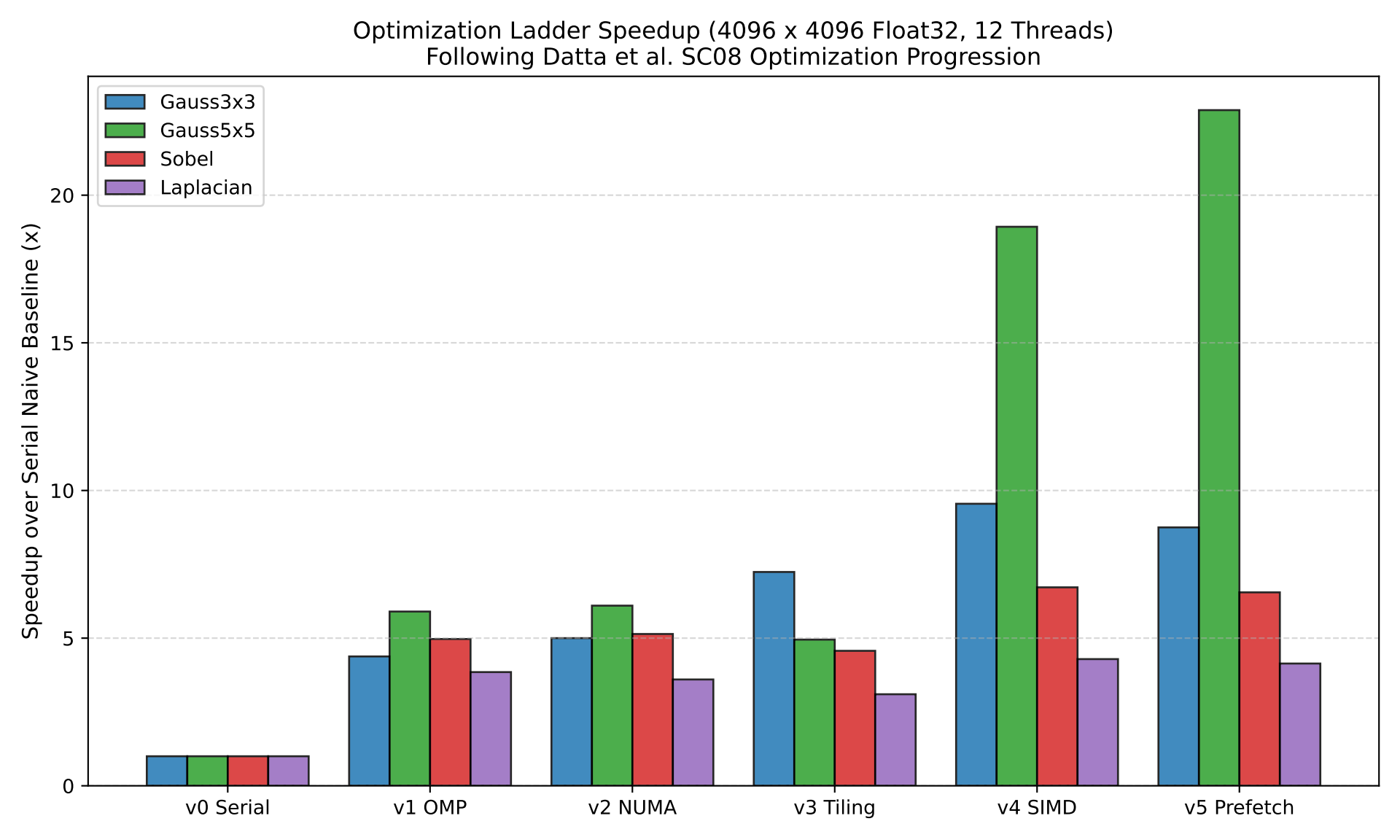
Optimization Ladder Speedup (4096 x 4096 Float32, 12 Threads)
Following Datta et al. SC08 Optimization Progression
Gauss3x3
Gauss5x5
Sobel
Laplacian
Speedup over Serial Naive Baseline (x)
0
5
10
15
20
v0 Serial
v1 OMP
v2 NUMA
v3 Tiling
v4 SIMD
v5 Prefetch